THE FOURIER ENTROPY–INFLUENCE INEQUALITY FOR THE CLIQUE PROPERTY

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Abstract. The Fourier entropy–influence conjecture of Friedgut and Kalai would give total influence $\Omega(\log^2 n)$ for every monotone graph property on n vertices with variance bounded below, and the property that $G(n,\frac{1}{2})$ contains a clique on about $2\log_2 n$ vertices shows that this would be sharp. We prove the conjecture for this property. If λ is the expected number of cliques on k vertices, then, uniformly over all k with $\lambda \le \Lambda$, the spectral entropy is at most $2+\log_2(e(5+4\lambda))+o(1)$ times the total influence as $n\to\infty$. The proof codes the set of vertices covered by the spectral sample, which for a graph property costs exactly the entropy of a uniformly random set of its size, and then the edges inside it as an arbitrary set of pairs. For monotone graph properties both costs are bounded by Parseval identities and the Harris inequality, and for cliques by Janson's inequality. The same argument applies to every increasing property of the graph induced on k vertices whose occurrences rarely overlap, with a bounded constant when $k=O(\log n)$, and in $G(n,p)$ for every $p\le\frac{1}{2}$ it gives the biased form of the conjecture proposed by Keller, Mossel and Schlank, with a constant that tends to 4 as $p\to 0$.

1. Introduction

For $f\colon\{0,1\}^N\to\{0,1\}$ let $F=1-2f$, $\chi_T(x)=(-1)^{\sum_{i\in T}x_i}$, $\widehat{F}(T)=\mathbb{E}[F\chi_T]$ with x uniform, and $P(T)=\widehat{F}(T)^2$. By Parseval P is a probability distribution on the subsets of $[N]$, and a random set S with law P is the *spectral sample* of f. The *spectral entropy* and the *total influence* are

$$H(f)=\sum_T P(T)\log_2\frac{1}{P(T)},\qquad I(f)=\mathbb{E}|S|=\sum_i \mathrm{Inf}_i(f),$$

where $\mathrm{Inf}_i(f)=\mathbb{P}[f(x)\neq f(x\oplus e_i)]=\mathbb{P}[i\in S]$. Friedgut and Kalai [5] conjectured that $H(f)\le C\,I(f)$ for a universal constant C. The conjecture is known for symmetric functions and read-once decision trees [15], read-once formulas [14], random linear threshold functions [4], functions of very small influence or very large entropy [17], δ-tribes [6], read-k DNF formulas and regular functions [18], and monotone DNF formulas of bounded width [19]; the best general bounds lose a factor that grows with the function [10, 11, 7]. It is open for monotone functions.

Take $N=\binom{n}{2}$ and identify $x\in\{0,1\}^N$ with a graph G on the vertex set $[n]$; for uniform x this is $G(n,\frac{1}{2})$. A *graph property* is a function invariant under permutations of the vertices. Friedgut and Kalai observed that their conjecture would give $I(f)\ge c\log^2 n$ for every monotone graph property whose variance is bounded below. Bourgain and Kalai [3] proved $I(f)\ge c_\delta\log^{2-\delta}n$, and the best bound known is $c\log^2 n/(\log\log n)^2$ [10]. The example that makes $\log^2 n$ sharp is the *clique property*

$$f_{n,k}(G)=\mathbf{1}[\,G\text{ contains a complete subgraph on }k\text{ vertices}\,].$$

Write $m=\binom{k}{2}$ and $\lambda=\lambda_{n,k}=\binom{n}{k}2^{-m}$ for the expected number of k-cliques. At the clique number of $G(n,\frac{1}{2})$ [2, 13], $k=2\log_2 n-2\log_2\log_2 n+O(1)$ and λ is of order one, and there

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$I(f_{n,k}) = (1+o(1))\,2\lambda m\,e^{-\lambda}$ (Lemma 4.1), which is of order $\log^2 n$, while the variance is bounded below.

The known bounds do not reach this function with a bounded constant. It is a monotone DNF formula whose $\binom{n}{k}$ terms have $m \sim 2\log_2^2 n$ variables each, and every variable lies in $\binom{n-2}{k-2}$ terms, so the bounded-width theorem of [19] gives the constant $m+8.1$ and the read-k theorem of [18] a constant of order $\log^2 n$. The bound $H(f) = O\big(I(f)+\sum_A P(A)|A|\log|A|\big)$ of [10] has a right-hand side of order at least $I\log I$, by Jensen's inequality, which is of order $I\log\log n$ here.

Theorem 1.1. *For every* $\Lambda > 0$,

$$\max_{2\le k\le n:\ \lambda_{n,k}\le\Lambda}\Big(\frac{H(f_{n,k})}{I(f_{n,k})} - 2 - \log_2\big(e(5+4\lambda_{n,k})\big)\Big) \le o(1) \qquad (n\to\infty).$$

In particular $\sup\{H(f_{n,k})/I(f_{n,k}) : 2\le k\le n,\ \lambda_{n,k}\le\Lambda\} < \infty$.

For $n > \Lambda$ the set $\{k \ge 1 : \lambda_{n,k} \le \Lambda\}$ consists of all k from the clique number, up to an additive constant, to n (Lemma 5.1). It contains the window in which $f_{n,k}$ has variance bounded below, where the constant is $6.61+o(1)$ for $\lambda \le 1$, and the whole range in which $f_{n,k}$ is close to 0, where it is $5.76+o(1)$ as $\lambda \to 0$. The second assertion follows from the first, because the finitely many functions with small n each have a finite ratio. Below the clique number $f_{n,k}$ is close to 1. There a per-coordinate estimate of [19], with the symmetry of graph properties and Janson's inequality, gives $H \le (8.96 + 2\log_2\binom{k}{2} + o(1))\,I$ uniformly for $3 \le k \le (2-\delta)\log_2 n$ (Corollary 7.2). At the values of n we computed, the same estimate stays below $8.96+2\log_2\binom{k}{2}$ for every k up to three below the clique number; at the last one or two values our bound is of order $\log n$ (Section 7).

Nothing in the argument needs $p = \frac12$. In Section 8 it is carried out in $G(n,p)$ for every $p \le \frac12$, where it proves the biased form $H_p(f) \le c\,p\log(1/p)\,I_p(f)$ of the conjecture, proposed by Keller, Mossel and Schlank [11], for the clique property on the same range of λ, uniformly in k and p (Theorem 8.3 and Corollary 8.4). As $p \to 0$ the constant tends to 4, which is the limit of the ratio for a dictator, so it cannot be improved.

The proof does not induct on coordinates. Proposition 2.1 bounds the entropy of any function of the edges of K_n by two costs: that of the set $V(S)$ of vertices covered by the edges of the spectral sample, coded by its size and then as one of the sets of that size, and that of S given $V(S)$, coded as an arbitrary set of pairs of $V(S)$. For a graph property the first cost cannot be lowered: the spectral distribution is invariant under permutations of the vertices, so $V(S)$ is uniformly distributed among the sets of its size and $H(f) \ge \mathbb{E}\log_2\binom{n}{|V(S)|}$, and the two costs characterize the conjecture for graph properties (Remark 9.2). For monotone graph properties the first cost is bounded through the probability that the spectral sample touches a given vertex, by a Parseval identity and the Harris inequality (Lemma 3.1), and the second by the number of pairs of vertices that the spectral sample covers, which a second-order Parseval identity controls in the same way (Lemma 3.2). For cliques Janson's inequality evaluates both. The spectral sample covers about k vertices and carries about $m/2$ of the pairs among them, so the vertex set costs about $k\log_2(n/k) \approx m$ bits and the edges a bounded number of bits each. The argument uses only that the pattern is dense and that its copies rarely share edges, and Theorem 4.4 states it for copies of an arbitrary graph. Theorem 6.1 extends it to any increasing property of the graph induced on k vertices, with the relative influence of the property in place of the number of edges of a copy; the edge-isoperimetric inequality bounds that influence from below and shows that the clique property is the extremal case, and the proof uses the Harris inequality in place of Janson's.

2. Vertex coding

Let f be a Boolean function of the edges of K_n, and let S be its spectral sample. Write $V(S)$ for the set of endpoints of the edges in S, $v = |V(S)|$, and $\varepsilon = \mathbb{P}[S \neq \emptyset] = 1 - \widehat{F}(\emptyset)^2$. For $W \subseteq [n]$ say that S *touches* W if some edge of S has an endpoint in W. Let h denote the binary entropy function.

Proposition 2.1. *If f is not constant, then*

$$H(f) \le A + \mathbb{E}\log_2 \binom{n}{v} + I \log_2 \frac{e\,\mathbb{E}\binom{v}{2}}{I}, \qquad A = h(\varepsilon) + \varepsilon \log_2 \frac{e\,\mathbb{E}v}{\varepsilon} + \varepsilon \log_2 \frac{eI}{\varepsilon},$$

where $I = I(f)$.

Proof. Since v and $V(S)$ are functions of S, $H(f) = H(S) = H(v) + H(V(S) \mid v) + H(S \mid V(S))$. A random variable with values in $\{1, 2, \dots\}$ and mean μ has entropy at most $\log_2(e\mu)$, the value of the geometric distribution being $\log_2 \mu + (\mu - 1)\log_2 \frac{\mu}{\mu - 1}$. The variable v vanishes exactly when S is empty and otherwise has conditional mean $\mathbb{E}v/\varepsilon$, so $H(v) \le h(\varepsilon) + \varepsilon \log_2(e\,\mathbb{E}v/\varepsilon)$. Given v, the set $V(S)$ is one of $\binom{n}{v}$ sets, so $H(V(S) \mid v) \le \mathbb{E}\log_2 \binom{n}{v}$.

Given $V(S)$, the set S is determined by $|S|$ and by its position among the $\binom{\binom{v}{2}}{|S|}$ sets of that size of pairs of $V(S)$. The variable $|S|$ vanishes exactly when $V(S)$ is empty, so in the same way $H(|S| \mid V(S)) \le \varepsilon \log_2(eI/\varepsilon)$. With $\log_2 \binom{M}{s} \le s \log_2(eM/s)$ and Jensen's inequality for the probability weights $P(T)|T|/I$,

$$\sum_T P(T) \log_2 \binom{\binom{v(T)}{2}}{|T|} \le \sum_T P(T)|T| \log_2 \frac{e\binom{v(T)}{2}}{|T|} \le I \log_2 \frac{e \sum_T P(T)\binom{v(T)}{2}}{I}. \qquad \square$$

3. Two estimates for monotone graph properties

Let f be a monotone function of the edges of K_n, and let G denote the random graph $G(n, \frac{1}{2})$. For $W \subseteq [n]$ let $G - W$ be G with every edge that has an endpoint in W deleted. A *minterm* of f is a minimal graph M with $f(M) = 1$; it *touches* W if it has an edge with an endpoint in W. For a set A of edges let $\mathbb{E}_A$ denote averaging over the edges in A.

Lemma 3.1. *For every $W \subseteq [n]$,*

$$\mathbb{P}[S \text{ touches } W] \le 4\,\mathbb{P}[f(G - W) = 0]\,\mathbb{P}[G \text{ contains a minterm of } f \text{ that touches } W].$$

Proof. Let A be the set of edges with an endpoint in W. The functions χ_T with $T \cap A \neq \emptyset$ span the orthogonal complement of the functions that do not depend on the edges in A, so $\mathbb{P}[S \text{ touches } W] = \sum_{T \cap A \neq \emptyset} \widehat{F}(T)^2 = \mathbb{E}\,(F - \mathbb{E}_A F)^2$. Fix the edges outside A. If $f(G - W) = 1$, then $f = 1$ for every choice of the edges in A, by monotonicity, and $F - \mathbb{E}_A F = 0$. Otherwise $\mathbb{E}_A(F - \mathbb{E}_A F)^2 = 4r(1 - r) \le 4r$, where r is the conditional probability that $f(G) = 1$. Therefore $\mathbb{P}[S \text{ touches } W] \le 4\mathbb{P}[f(G) = 1,\ f(G - W) = 0]$. If $f(G) = 1$ and $f(G - W) = 0$, then a minterm contained in G is not contained in $G - W$, so it touches W. The event that G contains a minterm touching W is increasing and $\{f(G - W) = 0\}$ is decreasing, and the Harris inequality [8] completes the proof. $\square$

Lemma 3.2. *Let $u \neq w$ be vertices, and let a, b, c, g be the indicators that G contains a minterm of f that touches u but not w, touches w but not u, touches both, and touches neither. Then*

$$\mathbb{P}[u, w \in V(S)] \le \mathbb{P}[uw \in S] + 8\big(\mathbb{E}[ab] + \mathbb{E}c\big)\mathbb{P}[g = 0].$$

Proof. If $u, w \in V(S)$ and $uw \notin S$, then S meets both A and B, where A and B are the sets of edges at u and at w other than uw. As in the previous proof, the functions χ_T with T meeting both A and B span the range of the commuting projections $(1-\mathbb{E}_A)(1-\mathbb{E}_B)$, so $\mathbb{P}[S \text{ meets } A \text{ and } B] = \|(1-\mathbb{E}_A)(1-\mathbb{E}_B)F\|_2^2 = 4\|(1-\mathbb{E}_A)(1-\mathbb{E}_B)f\|_2^2$. Fix the edges outside $A \cup B$; they include uw, and g depends on them only. If $g = 1$ then $f = 1$ on the whole fibre. Otherwise $f = 1-(1-a)(1-b)(1-c)$ on the fibre, a depends on the edges in A only and b on those in B only, and so

$$(1-\mathbb{E}_A)(1-\mathbb{E}_B)f = -(a-\mathbb{E}_A a)(b-\mathbb{E}_B b) + (1-\mathbb{E}_A)(1-\mathbb{E}_B)\big[(1-a)(1-b)c\big].$$

The operator $(1-\mathbb{E}_A)(1-\mathbb{E}_B)$ is a contraction and $\mathbb{E}_A(a-\mathbb{E}_A a)^2 \le \mathbb{E}_A a$, so the average of the square over the fibre is at most $2\mathbb{E}_A a\,\mathbb{E}_B b + 2\mathbb{E}_{A\cup B}\,c$. Hence $\mathbb{P}[S \text{ meets } A \text{ and } B] \le 8\,\mathbb{E}\big[(ab+c)\mathbf{1}[g=0]\big]$, and the Harris inequality applies to the increasing function $ab+c$ and the decreasing event $\{g=0\}$. □

Both lemmas were checked in exact arithmetic for every clique property on at most seven vertices, for every vertex, every pair of vertices, and the vertex sets obtained by halving $[n]$ recursively. The ratio of the two sides is at most 1 in Lemma 3.1, where it equals $\mathbb{P}[f=0]$ for $W=[n]$, and at most 0.39 in Lemma 3.2.

4. Copies of a graph

Let Γ be a graph without isolated vertices, with s vertices and m edges, let $\mathcal{K}$ be the set of subgraphs of K_n isomorphic to Γ (its *copies*), and let $f(G) = \mathbf{1}[G \supseteq K \text{ for some } K \in \mathcal{K}]$. The copies are the minterms of f. Put

$$\lambda = |\mathcal{K}|\,2^{-m}, \qquad \eta = \sum_{L \in \mathcal{K}\setminus\{K\}:\ E(L)\cap E(K)\neq\emptyset} 2^{-|E(L)\setminus E(K)|},$$

where $K \in \mathcal{K}$ is fixed; by symmetry η does not depend on K. Thus λ is the expected number of copies in G, and η the expected number of other copies that share an edge with a given copy, conditionally on its presence. For a set U of n' vertices the expected number of copies inside U is $\mu(n') = \lambda\binom{n'}{s}/\binom{n}{s}$, since the vertex set of a uniformly random copy is a uniformly random s-set. The sum over ordered pairs of distinct copies inside U that share an edge, of the probability that both are present, is at most $\lambda\eta$, and Janson's inequality [9], [1, Theorem 8.1.1] gives

$$\mathbb{P}[G \text{ contains no copy inside } U] \le \exp\big(-\mu(n') + \lambda\eta/2\big). \tag{1}$$

Also $1 - \binom{n-w}{s}/\binom{n}{s} \le sw/(n-s+1)$, so $\mu(n-w) \ge \lambda\big(1 - sw/(n-s+1)\big)$.

Lemma 4.1. *Let* $I_0 = 2\lambda m\big[\exp\big(-\lambda/(1-2^{-m}) - 2\eta\big) - \eta/2\big]$. *Then*

$$I_0 \le I(f) \le 2\lambda m \exp\Big(-\lambda + \frac{\lambda m}{\binom{n}{2}} + \frac{\lambda\eta}{2}\Big).$$

Proof. For an edge e let D be the event that $G-e$ contains no copy and, for a copy $K \ni e$, let A_K be the event that the edges of K other than e are present. The edge e is pivotal exactly on $D \cap \bigcup_{K\ni e} A_K$. Put $\mu_e = \sum_{K\ni e}\mathbb{P}[A_K]$; then $\sum_e \mu_e = m|\mathcal{K}|2^{1-m} = 2\lambda m$. The events A_K are increasing and D is decreasing, so $\mathrm{Inf}_e(f) \le \mu_e\mathbb{P}[D]$ by the Harris inequality. The copies not containing e have expected number $\lambda(1 - m/\binom{n}{2})$, the ordered pairs of them that share an edge have total probability at most $\lambda\eta$, and Janson's inequality for this family gives the upper bound. Conditionally on A_K the other edges are independent fair coins, and D is the intersection of the decreasing events that L is not present, over the copies L not containing e.

Such an L has $|E(L)\setminus E(K)|\ge 1$ edges outside K, and it is present with conditional probability $p_L = 2^{-|E(L)\setminus E(K)|}$. By the Harris inequality and $\log(1-p)\ge -p/(1-p)$,

$$\mathbb{P}[D \mid A_K] \ge \prod_L (1-p_L) \ge \exp\Big(-\frac{\lambda}{1-2^{-m}} - 2\eta\Big),$$

using $p_L = 2^{-m}$ for the at most $|\mathcal{K}|$ copies sharing no edge with K, and $p_L \le \frac12$ for the others, whose p_L sum to at most η. For distinct $K, K' \ni e$ the copy K' shares the edge e with K, so $\sum_{K'\ne K}\mathbb{P}[A_K\cap A_{K'}] \le \mathbb{P}[A_K]\,\eta$. By the Bonferroni inequality

$$\mathrm{Inf}_e(f) \ge \sum_{K\ni e}\mathbb{P}[A_K]\mathbb{P}[D\mid A_K] - \tfrac12\sum_{K\ne K'\ni e}\mathbb{P}[A_K\cap A_{K'}] \ge \mu_e\Big[\exp\Big(-\frac{\lambda}{1-2^{-m}}-2\eta\Big) - \frac{\eta}{2}\Big],$$

and summing over e gives the lower bound. □

Lemma 4.2. $\mathbb{E}v \le 4\lambda s\exp\big(-\lambda+\lambda s/(n-s+1)+\lambda\eta/2\big)$ *and* $\varepsilon \le \min(1,4\lambda)$.

Proof. $\mathbb{E}v = \sum_u \mathbb{P}[S \text{ touches } \{u\}]$. By Lemma 3.1, with $f(G-u)=0$ meaning that there is no copy inside $[n]\setminus\{u\}$, and by (1), each term is at most $4\,(\lambda s/n)\,e^{-\mu(n-1)+\lambda\eta/2}$, because the expected number of copies containing u is $\lambda s/n$; and $\mu(n-1)\ge\lambda(1-s/(n-s+1))$. Finally $\varepsilon = 4q(1-q)$ with $q=\mathbb{P}[f=0]$, and $1-q\le\lambda$. □

Lemma 4.3.

$$\mathbb{E}\binom{v}{2} \le I(f) + 4\big(\lambda^2s^2+\lambda s^2\eta+\lambda s(s-1)\big)\exp\Big(-\lambda+\frac{2\lambda s}{n-s+1}+\frac{\lambda\eta}{2}\Big).$$

Proof. Sum Lemma 3.2 over the ordered pairs (u,w); the terms $\mathbb{P}[uw\in S]$ add up to $2I$, and $\mathbb{E}\binom{v}{2}$ is half the sum. A copy touching both u and w contains both, so $\sum_{u\ne w}\mathbb{E}c \le \lambda s(s-1)$. For $\mathbb{E}[ab]$ count pairs (K,K') of copies with $u\in V(K)\setminus V(K')$ and $w\in V(K')\setminus V(K)$. Pairs that share no edge are present with probability 4^{-m}, and they contribute at most $\big(\lambda s(n-s)/(n(n-1))\big)^2$ for each (u,w), hence at most λ^2s^2 in total. A pair that shares an edge is counted for at most s^2 pairs (u,w), and the ordered pairs of copies sharing an edge have total probability at most $\lambda\eta$. Finally $\{g=0\}$ is the event that there is no copy inside $[n]\setminus\{u,w\}$, and (1) applies with $\mu(n-2)\ge\lambda(1-2s/(n-s+1))$. □

Theorem 4.4. *Let Γ_n be graphs without isolated vertices, with s_n vertices and m_n edges, such that $s_n\to\infty$, $s_n/n\to 0$, $(s_n\log s_n)/m_n\to 0$, $\lambda_n\le\Lambda$ and $\eta_n\to 0$, and let $\rho_n = \binom{s_n}{2}/m_n$. Then*

$$H(f_n) \le \big(2+\log_2\big(e(1+4\rho_n(1+\lambda_n))\big)+o(1)\big)\,I(f_n).$$

The $o(1)$ depends only on Λ and on upper bounds for $1/s_n$, s_n/n, $(s_n\log s_n)/m_n$ and η_n.

For dense patterns the third condition is automatic: it holds whenever $\rho_n = o(s_n/\log s_n)$.

Proof. Drop the index n. Put $x=\log_2|\mathcal{K}|/m$, so that $\log_2(1/\lambda)=m(1-x)$ and $x\le 1+\log_2\Lambda/m$. Since $|\mathcal{K}|\ge\binom{n}{s}\ge(n/s)^s$,

$$s\log_2(n/s) \le xm. \tag{2}$$

By Lemma 4.1, $I\ge I_0 = 2\lambda m e^{-\lambda}(1-o(1))$. Since $v\ge 2$ whenever $v>0$, $\log_2\binom{n}{v}\le v\log_2(en/2)$, so by Lemma 4.2 and (2) the vertex term $V=\mathbb{E}\log_2\binom{n}{v}$ satisfies

$$\frac{V}{I} \le \frac{4\lambda s\log_2(en/2)(1+o(1))}{2\lambda m} \le \Big(2x+\frac{2s\log_2(es/2)}{m}\Big)(1+o(1)) = 2x+o(1).$$

Consider A from Proposition 2.1. Its middle term is small: the map $y\mapsto y\log_2(ec/y)$ increases for $y\le c$, $\varepsilon\le\mathbb{E}v$ and $\varepsilon\le\min(1,4\lambda)$, so by Lemma 4.2 the term is at most $4\lambda\log_2(es\,e^{\lambda\eta/2+o(1)})$,

which is $o(I)$ because $I \ge I_0$ and $\log s = o(m)$. Let $A_0 = h(\varepsilon) + \varepsilon \log_2(eI/\varepsilon)$ be the rest. If $\lambda \ge \frac14$ then $A_0 \le 2 + \log_2(eI)$ and $I \ge I_0 \to \infty$, so $A_0/I = o(1)$. If $\lambda < \frac14$, then $\varepsilon \le 4\lambda < 1$ and $h(\varepsilon) \le \varepsilon \log_2(e/\varepsilon)$, so $A_0 \le \phi(\varepsilon)$ with $\phi(y) = y \log_2(e^2 I/y^2)$, which increases for $y \le \sqrt{I}$; since $I \ge I_0 \ge 16\lambda^2$, $A_0 \le \phi(4\lambda)$. The function $I \mapsto \phi(4\lambda)/I$ decreases for $I \ge 16\lambda^2/e$, so

$$\frac{A_0}{I} \le \frac{4\lambda}{I_0} \log_2 \frac{e^2 I_0}{16\lambda^2} \le \frac{2e^\lambda(1+o(1))}{m}\Big(m(1-x) + \log_2 \frac{e^2 m}{8}\Big) = 2e^\lambda(1-x)(1+o(1)) + o(1).$$

In both cases $A/I + V/I \le 2 + 2(e^\lambda - 1)(1-x)^+ + o(1)$, and for $\lambda < \frac14$, $(e^\lambda - 1)(1-x) = (e^\lambda - 1)\log_2(1/\lambda)/m \le 1/m$. Hence $A/I + V/I \le 2 + o(1)$.

By Lemmas 4.1 and 4.3, and $s^2/m = 2\rho s/(s-1)$,

$$\frac{\mathbb{E}\binom{v}{2}}{I} \le 1 + \frac{4(\lambda s^2 + s^2\eta + s(s-1))(1+o(1))}{2m} \le \big(1 + 4\rho(1+\lambda)\big)(1+o(1)).$$

Proposition 2.1 now gives the theorem. □

5. The clique property

For $\Gamma = K_k$ we have $s = k$, $m = \binom{k}{2}$, $|\mathcal{K}| = \binom{n}{k}$ and $\rho = 1$, and

$$\eta = \sum_{j=2}^{k-1} t_j, \qquad t_j = \binom{k}{j}\binom{n-k}{k-j} 2^{-(m - \binom{j}{2})},$$

since a k-clique meeting a given one in j vertices shares $\binom{j}{2}$ edges with it.

Lemma 5.1. *Let $\Lambda > 0$. For $n > \Lambda$ the set $\{1 \le k \le n : \lambda_{n,k} \le \Lambda\}$ is an interval $\{k_1, \dots, n\}$ with $k_1 = 2\log_2 n - 2\log_2\log_2 n + O(1)$. Moreover $\eta_{n,k} \le \eta^*(n)$ for all k in this interval, where $\eta^*(n) \to 0$.*

Proof. The ratio $\lambda_{n,k+1}/\lambda_{n,k} = \frac{n-k}{k+1} 2^{-k}$ decreases in k, so $\lambda_{n,k}$ increases and then decreases in k, and $\lambda_{n,1} = n > \Lambda$; hence the set is an interval ending at n. If $\lambda_{n,k} \le \Lambda$, then $(n/k)^k \le \binom{n}{k} \le \Lambda 2^{k(k-1)/2}$, that is,

$$2^{(k-1)/2} \ge \frac{n}{k} \Lambda^{-1/k}, \tag{3}$$

which gives $k \ge 2\log_2 n - 2\log_2\log_2 n - O(1)$. Conversely $\lambda_{n,k} \le \big(e\, n\, 2^{-(k-1)/2}/k\big)^k$, which is at most Λ once $2^{(k-1)/2} \ge 2en/k$ and $k \ge \log_2(1/\Lambda)$; this holds for $k \ge 2\log_2 n - 2\log_2\log_2 n + C_\Lambda$.

For η split the sum at $j = k/2$. If $j \ge k/2$ write $i = k - j \in [1, k/2]$; then $m - \binom{j}{2} = i(2k-i-1)/2 \ge i(3k/2-1)/2$ and $\binom{k}{i}\binom{n-k}{i} \le (kn)^i$, so $t_{k-i} \le y^i$ with $y = \sqrt{2}\, kn\, 2^{-3k/4}$. By (3), $y \le 2^{-1/4} k^{5/2} \Lambda^{3/(2k)} n^{-1/2}$, and for $k > 3\log_2 n$ directly $y \le \sqrt{2}\, k\, n^{-5/4}$; in both cases $y \to 0$ uniformly. If $2 \le j \le k/2$, then $\binom{k}{j}\binom{n-k}{k-j}/\binom{n}{k} \le \binom{k}{j}\big(k/(n-k+1)\big)^j$, so $t_j \le \lambda\big(k^2 2^{(j-1)/2}/(n-k+1)\big)^j/j!$. For $k \le 3\log_2 n$ this is at most Λz^j with $z = 2k^2 n^{-1/4} \to 0$. For $k > 3\log_2 n$ use instead $\binom{k}{j} \le 2^k$, $\binom{n-k}{k-j} \le n^{k-j}$ and $m - \binom{j}{2} = (k-j)(k+j-1)/2 \ge (k-j)(k+1)/2$: then $t_j \le 2^k (n 2^{-(k+1)/2})^{k-j} \le (2n^{-1/4})^k$. Altogether $\eta \le y/(1-y) + 2\Lambda z^2 + k(2n^{-1/4})^k \to 0$. □

Proof of Theorem 1.1. By Lemma 5.1, the k with $\lambda_{n,k} \le \Lambda$ satisfy $k \to \infty$ and $\eta \to 0$ uniformly. For $k \le n/\log_2 n$ the proof of Theorem 4.4 applies uniformly and gives the bound with $\rho = 1$.

Let $k > n/\log_2 n$. Then $\lambda \le 2^n 2^{-k(k-1)/2}$, so $\lambda n \to 0$. By Lemma 3.1 every vertex is touched with probability at most $4\lambda k/n$, so $\mathbb{E}v \le 4\lambda k$, and $V \le \mathbb{E}v \log_2(en/2)$ gives $V/I \le 4\log_2(en/2)/(k-1)\,(1+o(1)) \to 0$. The estimate of A/I in the proof of Theorem 4.4 used only $\varepsilon \le 4\lambda$, $\mathbb{E}v = O(\lambda k)$ and $I \ge I_0$, so $A/I \le 2 + o(1)$. Since $\lambda n \to 0$, Lemmas 4.1 and 4.3 give $\mathbb{E}\binom{v}{2}/I \le 5 + o(1)$. Hence $H/I \le 2 + \log_2(5e) + o(1)$, which is at most the bound claimed. □

6. Local properties

The proof of Theorem 4.4 uses the pattern only through λ, the overlap η, and the lower bound $I \ge 2\lambda m e^{-\lambda}(1-o(1))$, where $2m$ is the total influence of a copy divided by its probability. Here the copies are replaced by an arbitrary increasing property of the graph induced on a k-set. The role of $2m$ is taken by the relative influence of that property, which the edge-isoperimetric inequality bounds from below, and the probabilities of $\{f=0\}$ enter only as ratios, which the Harris inequality controls, so Janson's inequality is not needed.

Let B be an increasing property of graphs on $[k]$, invariant under relabelling, whose minimal elements have no isolated vertex. For a k-set $W \subseteq [n]$ let B_W be the event that the graph induced by G on W lies in B, and let $f(G) = \mathbf{1}[B_W \text{ for some } W]$. Put

$$\pi = \mathbb{P}\big[G(k,\tfrac12) \in B\big], \qquad \lambda = \binom{n}{k}\pi, \qquad \iota = \frac{I(B)}{\pi},$$

where $I(B)$ is the total influence of B as a function of the $\binom{k}{2}$ edges. For $2 \le j \le k-1$ let π_j be the conditional probability of B given that all edges inside a fixed j-subset of $[k]$ are present, and

$$\eta = \sum_{j=2}^{k-1} \binom{k}{j}\binom{n-k}{k-j}\pi_j.$$

For $B = \{K_k\}$ we have $\pi = 2^{-m}$, $\pi_j = 2^{-(m-\binom{j}{2})}$, $\iota = 2m$, and η is that of Section 5. By the edge-isoperimetric inequality of the cube, $I(B) \ge 2\pi \log_2(1/\pi)$ for $\pi \le \frac12$, so

$$\iota \ge 2\log_2 \frac{1}{\pi} = 2\log_2 \frac{\binom{n}{k}}{\lambda}, \tag{4}$$

with equality for $B = \{K_k\}$, because a subcube attains the edge-isoperimetric inequality.

Theorem 6.1. *Let B_n be properties as above, on k_n vertices, with $\lambda_n \le \Lambda$, $\eta_n \to 0$ and $k_n \log k_n = o(\log(1/\pi_n))$. Then*

$$H(f_n) \le \Big(2 + \log_2\Big(e\Big(1 + \frac{4(1+\lambda_n)k_n^2}{\iota_n}\Big)\Big) + o(1)\Big) I(f_n).$$

By (4), $4k^2/\iota \le 2k^2/\log_2(1/\pi)$, so the constant is bounded when $\log_2(1/\pi)$ is of order k^2. Since $\log_2(1/\pi) = \log_2\binom{n}{k} - \log_2\lambda \ge k\log_2(n/k) - \log_2\Lambda$, the bound is at most $2 + \log_2\big(e(1 + 2(1+\lambda)k/\log_2(n/k))\big) + o(1)$ when $\log k = o(\log n)$. The clique property is the extremal case: by (4), at the same n, k and λ every other B has a larger ι and a smaller bound. For $B = \{K_k\}$ we have $\iota = 2m$, $\log_2(1/\pi) = m$ and $\eta \to 0$ by Lemma 5.1, and Theorem 6.1 gives the bound of Theorem 1.1 again, with a proof that does not use Janson's inequality. Since B contains the complete graph, $\lambda \ge \lambda_{n,k}$, so $k \ge k_1(n) = 2\log_2 n - O(\log\log n)$ by Lemma 5.1.

Proof. Drop the index n, write $q = \mathbb{P}[f=0]$ and $L = \log_2(1/\pi) = \log_2\big(\binom{n}{k}/\lambda\big)$, and let θ stand for a nonnegative quantity that tends to 0, not the same at each occurrence. Since $\lambda \ge \lambda_{n,k}$, Lemma 5.1 gives $k \to \infty$, so $L \to \infty$ and $k\log k = o(L)$. Moreover $\lambda k/n \to 0$ and $\lambda k^4/n^2 \to 0$. This is clear if $k \le n^{1/3}$. Otherwise $\log_2\binom{n}{k} \le k\log_2(en/k) \le k(2\log_2 k + \log_2 e) = o(L)$, so $\log_2(1/\lambda) = L - \log_2\binom{n}{k} \ge L/2$, while $L/\log_2 n \to \infty$.

Ratios. For a set U of at most two vertices, $\{f=0\}$ is the intersection of $\{f(G-U)=0\}$ with the event that B_W fails for every W meeting U. Both events are decreasing, so by the Harris inequality, applied twice,

$$\mathbb{P}[f(G-U)=0] \le \frac{q}{(1-\pi)^{|U|\binom{n-1}{k-1}}} \le q\exp\Big(\frac{|U|\lambda k}{n(1-\pi)}\Big) = q(1+\theta). \tag{5}$$

Here we used that $f(G-U)=0$ is the event that B_W fails for every W disjoint from U, because a minimal element of B has no isolated vertex. For the same reason, the minterms of f that touch a vertex u lie in the events B_W with $W \ni u$.

Influence. Fix an edge e and a k-set $W \ni e$. Let P_W be the event that e is pivotal for B_W, which depends only on the edges inside W other than e, and let D be the event that $f(G-e)=0$. The edge e is pivotal for f exactly on $D \cap \bigcup_{W \ni e} P_W$. Condition on a configuration ξ of the edges inside W in which P_W holds. The event D is the intersection of the events $\neg B_{W'}$ over the k-sets $W' \neq W$ with $|W \cap W'| \geq 2$ (the near sets) and of the event D_{far} that $B_{W'}$ fails for every W' with $|W \cap W'| \leq 1$, which does not depend on ξ. All these are decreasing in the edges outside W, and $\mathbb{P}[B_{W'} \mid \xi] \leq \pi_{|W \cap W'|}$ by monotonicity. Hence, by the Harris inequality and since every $\pi_j \leq \eta \leq \frac{1}{2}$ for large n,

$$\mathbb{P}[P_W \cap D] \geq \mathbb{P}[P_W]\, e^{-2\eta} \mathbb{P}[D_{\mathrm{far}}] \geq \mathbb{P}[P_W]\, e^{-2\eta} q.$$

For $W' \ni e$ with $W' \neq W$, in the same way $\mathbb{P}[P_W \cap P_{W'} \cap D] \leq \mathbb{P}[P_W]\, \pi_{|W \cap W'|} \mathbb{P}[D_{\mathrm{far}}]$. The expected number of near sets W' with $B_{W'}$ is at most $\lambda \binom{k}{2}^2 / \binom{n}{2} = \theta$, so $q \geq \mathbb{P}[D_{\mathrm{far}}](1-\pi-\theta)$ by the Harris inequality and the union bound, and $\mathbb{P}[D_{\mathrm{far}}] \leq q(1+\theta)$. By the Bonferroni inequality and summation over e and W,

$$I(f) \geq q \binom{n}{k} I(B) \Big(e^{-2\eta} - \frac{\eta}{2}(1+\theta) \Big) = q\lambda\iota(1-\theta). \tag{6}$$

Vertices and pairs. By Lemma 3.1 and (5), every vertex is touched with probability at most $4q(1+\theta)\lambda k/n$, so $\mathbb{E}v \leq 4q\lambda k(1+\theta)$. By Lemma 3.2 and (5), and by the counting in the proof of Lemma 4.3, with $\mathbb{P}[B_W \cap B_{W'}] \leq \pi\, \pi_{|W \cap W'|}$ when $|W \cap W'| \geq 2$,

$$\mathbb{E}\binom{v}{2} \leq I + 4q(1+\theta)\big(\lambda^2 k^2 + \lambda k^2 \eta + \lambda k(k-1)\big).$$

With (6) this gives $\mathbb{E}\binom{v}{2}/I \leq \big(1 + 4(1+\lambda)k^2/\iota\big)(1+\theta)$.

The terms A and V. As in the proof of Theorem 4.4, $V \leq \mathbb{E}v \log_2(en/2)$. Since $k \log_2(en/2) = k \log_2(n/k) + k \log_2(ek/2) \leq L + \log_2 \lambda + k \log_2(ek/2)$, (6) and (4) give

$$\frac{V}{I} \leq \frac{4k \log_2(en/2)}{\iota}(1+\theta) \leq \frac{2\big(L + \log_2 \lambda + k \log_2(ek/2)\big)}{L}(1+\theta).$$

The middle term of A is at most $\mathbb{E}v \log_2 e$, since $y \log_2(ec/y) \leq c \log_2 e$, and this is $O(kI/\iota) = o(I)$. If $\lambda \geq \frac{1}{4}$, then $q \geq (1-\pi)^{\binom{n}{k}} \geq e^{-2\Lambda}$ by the Harris inequality, so $I \geq e^{-2\Lambda}\iota/5 \to \infty$ and $A_0/I = o(1)$, while $V/I \leq 2+\theta$ because $\log_2 \lambda \leq \log_2 \Lambda$ and $k \log_2(ek/2) = o(L)$. If $\lambda < \frac{1}{4}$, then $q \geq 1-\lambda$, and as in the proof of Theorem 4.4

$$\frac{A_0}{I} \leq \frac{4\lambda}{I} \log_2 \frac{e^2 I}{16\lambda^2} \leq \frac{4 \log_2(1/\lambda)}{\iota}(1+\theta) + \theta,$$

so that

$$\frac{A_0 + V}{I} \leq \frac{2\big(\log_2(1/\lambda) + L + \log_2 \lambda + k \log_2(ek/2)\big)}{L}(1+\theta) + \theta = 2 + \theta.$$

Proposition 2.1 now gives the theorem. □

For $t \geq 0$ call a set W of vertices *t-defective* if every vertex of W is adjacent to all but at most t of the other vertices of W. For $t = 0$ these are the cliques; in network analysis they are the $(t+1)$-plexes of [16]. Let $f^{(t)}_{n,k}(G)$ be the indicator that G has a t-defective set of k vertices, and $\lambda^{(t)}_{n,k}$ the expected number of such sets.

Corollary 6.2. *Let $\Lambda > 0$ and let $t_n \log\log n = o(\log n)$. Uniformly over $0 \le t \le t_n$ and over the k with $\lambda_{n,k}^{(t)} \le \Lambda$,*

$$H\big(f_{n,k}^{(t)}\big) \le \Big(2 + \log_2\big(e(5 + 4\lambda_{n,k}^{(t)})\big) + o(1)\Big) I\big(f_{n,k}^{(t)}\big).$$

Proof. Here B consists of the graphs on $[k]$ whose complement has maximum degree at most t. Since $\lambda_{n,k}^{(t)} \ge \lambda_{n,k}$, Lemma 5.1 gives $k \ge k_1(n)$, so every graph in B has minimum degree $k-1-t \ge 1$, and $t\log k = o(k)$. A graph on $[k]$ of maximum degree at most t is determined by the neighbourhoods of its vertices, so there are at most $N \le (1+k^t)^k$ of them, and $\pi = 2^{-m}N$. Hence $\log_2(1/\pi) \ge m - k\log_2(1+k^t) = m(1-\theta)$, and by (4), $4(1+\lambda)k^2/\iota \le 4(1+\lambda)(1+\theta)$. It remains to show that $\eta \to 0$ uniformly; then Theorem 6.1 gives the bound.

Write $t_j = \binom{k}{j}\binom{n-k}{k-j}\pi_j$. For $j \le k/2$ we have $\pi_j \le 2^{\binom{j}{2}}\pi$, because $\pi \ge 2^{-\binom{j}{2}}\pi_j$. So if $k \le 3\log_2 n$, the bound for $j \le k/2$ in the proof of Lemma 5.1 holds with $\lambda_{n,k}^{(t)} \le \Lambda$ in place of λ. If $k > 3\log_2 n$, then $\pi_j \le 2^{-(m-\binom{j}{2})}N$ and $m - \binom{j}{2} \ge 3k^2/8 - k/2$ give $t_j \le 2^k n^k 2^{-3k^2/8+k/2}(1+k^t)^k = 2^{-\Omega(k^2)}$. For $j = k-i$ with $1 \le i \le k/2$, a graph of maximum degree at most t with no edge inside a fixed $(k-i)$-set is determined by the neighbourhoods of the other i vertices, so $\pi_{k-i} \le 2^{-i(2k-i-1)/2}(1+k^t)^i$ and

$$t_{k-i} \le \big(kn(1+k^t)2^{-(3k-2)/4}\big)^i.$$

If $k \le 3\log_2 n$, then (3) holds, because $\lambda_{n,k} \le \lambda_{n,k}^{(t)} \le \Lambda$, and it bounds the base by

$$2(1+k^t)\,k^{5/2}\Lambda^{3/(2k)}n^{-1/2},$$

which tends to 0 because $t\log k = o(\log n)$. If $k > 3\log_2 n$, the base is at most $8k^{t+1}n\,2^{-3k/4}$, which tends to 0. Hence $\eta \to 0$ uniformly. □

7. Below the clique number

For k below the clique number, $\lambda \to \infty$ and $f_{n,k}$ is close to 1. The bound of Theorem 4.4 then grows like $\log_2\lambda$, and its Janson estimates fail once $\lambda\eta$ is large. A bound that grows only like $\log k$ follows from the per-coordinate estimate of [19], the symmetry of graph properties, the edge-isoperimetric inequality and Janson's inequality. Let $\Psi(r) = \log_2\big(e(r + 2\sqrt{r} + 2)\big)$.

Proposition 7.1. *Let f be a monotone graph property on n vertices, let $N = \binom{n}{2}$ and $\mu = \min(\mathbb{P}[f=0], \mathbb{P}[f=1]) > 0$. Then*

$$H(f) \le \Big(4.6355 + \Psi\Big(\frac{2N}{\log_2(1/\mu)} - 2\Big)\Big)\, I(f).$$

Proof. By [19, Corollary 1.4], every monotone f satisfies

$$H(f) \le \sum_e \mathrm{Inf}_e(f)\Big(4.6355 + \Psi\Big(\frac{4\mu}{\mathrm{Inf}_e(f)} - 2\Big)\Big).$$

For a graph property every $\mathrm{Inf}_e(f)$ equals $I(f)/N$. By the edge-isoperimetric inequality of the cube, applied to the smaller of the sets $\{f=0\}$ and $\{f=1\}$, $I(f) \ge 2\mu\log_2(1/\mu)$. Hence $4\mu N/I(f) \le 2N/\log_2(1/\mu)$, and Ψ is increasing. □

Corollary 7.2. *For all $3 \le k \le n$,*

$$\frac{H(f_{n,k})}{I(f_{n,k})} \le 4.6355 + \Psi\Big(4\ln 2 \cdot \frac{N\max(1,\eta)}{\lambda} - 2\Big).$$

For every $\delta > 0$, uniformly over $3 \le k \le (2-\delta)\log_2 n$,

$$H(f_{n,k}) \le \Big(4.6355 + \Psi\big(8\ln 2\,\tbinom{k}{2}^2\big) + o(1)\Big) I(f_{n,k}) \le \Big(8.96 + 2\log_2\tbinom{k}{2} + o(1)\Big) I(f_{n,k}).$$

Proof. Janson's inequality gives $\mathbb{P}[f=0] \le \exp(-\lambda+\lambda\eta/2)$, and $\mathbb{P}[f=0] \le \exp(-\lambda/(2\eta))$ when $\eta \ge 1$ [1, Theorem 8.1.2]. In both cases $\ln(1/\mu) \ge \lambda/(2\max(1,\eta))$, and Proposition 7.1 gives the first bound.

For the second, write $t_j = \lambda\rho_j 2^{\binom{j}{2}}$ with $\rho_j = \binom{k}{j}\binom{n-k}{k-j}/\binom{n}{k}$. As in the proof of Lemma 5.1, $\rho_j \le \binom{k}{j}\big(k/(n-k+1)\big)^j$, while $\rho_2 \ge \binom{k}{2}\frac{k(k-1)}{n(n-1)}\big(1-O(k^2/n)\big)$. Since $\binom{j}{2} = 1 + (j-2)(j+1)/2$, this gives, for $3 \le j \le k-1$,

$$\frac{t_j}{t_2} \le \frac{2(1+o(1))}{j!}\Big(\frac{k}{k-1}\Big)^2\Big(\frac{k^2\,2^{(j+1)/2}}{n-k+1}\Big)^{j-2} \le \frac{5}{j!}\,z^{j-2}, \qquad z = \frac{k^2\,2^{k/2}}{n-k+1} \le \frac{\sqrt{2}\,k^2}{n^{\delta/2}}(1+o(1)),$$

because $2^{(k-1)/2} \le n^{1-\delta/2}$; for $k=3$ there is no such j. So $\eta = (1+o(1))t_2$. Moreover $t_2 \to \infty$: for $k > 8/\delta$, $\lambda \ge (n/k)^k 2^{-k(k-1)/2} \ge (n^{\delta/2}/k)^k \ge n^{\delta k/4}$ and $t_2 \ge \lambda k^2(k-1)^2/(2n^2)$, while for bounded k the expected number t_2 is of order n^{k-2}. Hence $\max(1,\eta) = \eta$ for large n and

$$\frac{N\max(1,\eta)}{\lambda} = (1+o(1))\,2N\rho_2 = (1+o(1))\,2\binom{k}{2}^2.$$

Finally $\Psi(8\ln 2\,x^2) = \log_2(8e\ln 2) + 2\log_2 x + \log_2\big(1+0.85/x+0.37/x^2\big) \le 3.915 + 2\log_2 x + 0.405$ for $x = \binom{k}{2} \ge 3$. □

The loss $2\log_2\binom{k}{2} \approx 4\log_2 k$ comes from the ratio $N/\log_2(1/\mu)$, that is, from Janson's lower-tail exponent $\lambda/(2\eta)$, which is of order n^2/k^4. The true value of $\log_2(1/\mathbb{P}[f=0])$ is at most $(1+o(1))N/(k-1)$, because a $(k-1)$-partite graph has no K_k, so this route cannot lose less than $\log_2(k-1)$. For fixed k the width bound $\binom{k}{2} + 8.1$ of [19] is better when $k \le 4$ and worse from $k=5$ on.

The first bound of Corollary 7.2 covers the range between $(2-\delta)\log_2 n$ and the clique number as well, with a constant that grows as λ decreases. Together with Remark 9.1 it gives, for each of $n = 2^{12}, 2^{16}, 2^{20}, 2^{24}, 2^{32}, 2^{40}$ and every k from 3 to the first k_1 with $\lambda_{n,k_1} \le 11$, a bound for H/I. Taking the smallest of the bound of Corollary 7.2, the computed bound $B(n,k)$ of Remark 9.1 and the width bound, the largest value over these k is 23.2, 25.0, 31.7, 33.7, 38.4 and 34.0 respectively. It is attained at k_1-1 or k_1-2. There λ lies between Λ and a small power of n: the factor $e^{\lambda\eta/2}$ in (1) is large, so the estimates of Section 4 are weak, and Corollary 7.2 loses about $\log_2(N/\lambda)$. For $6 \le k \le k_1 - 3$ the bound is within 0.08 of $8.55 + 2\log_2\binom{k}{2}$, its limit when t_2 dominates η and $k \to \infty$, and for $3 \le k \le 5$ within 0.35.

Remark 7.3 (The pieces of the spectral sample). The following heuristic suggests that for $\lambda \to \infty$ the loss is in the coding and not in the estimates. As long as copies rarely overlap, $1-f$ is close to a product of the factors $1 - \mathbf{1}[K \subseteq G]$. For a product of functions of disjoint sets of variables the normalized spectral distribution is the product of the factors' distributions, and for one factor it is the empty set with probability $1-2^{-m}$ and otherwise a uniformly random nonempty set of edges of K. The spectral sample of f, given that it is nonempty, would then be a union of a Poisson(λ) number of such pieces, with $\mathbb{E}\binom{v}{2}$ about $\lambda^2k^2/2$ and $\mathbb{E}|S|$ about $\lambda k^2/4$ on that event. Coding S inside $V(S)$ as an arbitrary set of pairs then pays about $\log_2\lambda$ bits per edge, where naming the piece of each vertex would cost $\log_2\lambda$ bits per vertex. A proof along these lines needs a Fourier-side handle on the pieces, and none is known to us. For fixed k and $n \to \infty$, almost every K_k-free graph is $(k-1)$-partite [12]; heuristically the spectral sample of $1-f$ is then a random set of pairs inside the parts of a random partition into $k-1$ parts, which covers all vertices and has $\mathbb{E}\binom{v}{2}$ of order kI.

8. Biased measures

Let $0 < p \le \frac12$ and let μ_p be the product measure on $\{0,1\}^N$ under which each coordinate equals 1 with probability p; for $N = \binom{n}{2}$ it is the law of $G(n,p)$. Put $\sigma = \sqrt{p(1-p)}$ and $\phi_T(x) = \prod_{i\in T}(x_i - p)/\sigma$. These functions form an orthonormal basis of $L^2(\mu_p)$, and $\widehat{F}(T) = \mathbb{E}_{\mu_p}[F\phi_T]$, $P(T) = \widehat{F}(T)^2$ define a spectral sample S, with entropy $H_p(f)$ and mean size $I_p^{\mathrm{s}}(f) = \mathbb{E}|S|$. Conditionally on the other coordinates, F has variance $4p(1-p)$ in a coordinate that is pivotal and 0 in one that is not, so $\mathbb{P}[i \in S] = 4p(1-p)\mathbb{P}_p[i \text{ is pivotal}]$ and

$$I_p^{\mathrm{s}}(f) = 4p(1-p)\, I_p(f), \qquad I_p(f) = \sum_i \mathbb{P}_p\big[f(x) \ne f(x \oplus e_i)\big].$$

Keller, Mossel and Schlank [11, Conjecture 1.6] conjectured that $H_p(f) \le c\, p\log(1/p)\, I_p(f)$ for every f; the ratio of the two sides does not depend on the base of the logarithms. They showed that this follows from the conjecture for the uniform measure, that the property of containing K_r, for fixed r at its threshold, makes the factor $p\log(1/p)$ necessary, and that the bound holds with $\log N$ in place of $\log(1/p)$, which for the clique property settles the case of bounded k, with a constant depending on k.

Sections 2–4 carry over to μ_p, with I_p^{s} in place of I. Proposition 2.1 concerns an arbitrary random set of edges. In Lemmas 3.1 and 3.2 the identities $\mathbb{P}[S \text{ touches } W] = \mathbb{E}(F - \mathbb{E}_A F)^2$ and $\mathbb{P}[S \text{ meets } A \text{ and } B] = \|(1-\mathbb{E}_A)(1-\mathbb{E}_B)F\|_2^2$ hold for any product measure, with $\mathbb{E}_A$ the average over the coordinates in A under μ_p, and so does the Harris inequality. In Section 4 put

$$\lambda = |\mathcal{K}|\, p^m, \qquad \eta = \sum_{L \in \mathcal{K}\setminus\{K\}:\ E(L)\cap E(K)\ne\emptyset} p^{|E(L)\setminus E(K)|}.$$

Janson's inequality (1) holds under μ_p, and so do Lemmas 4.2 and 4.3, with the same proofs. In the proof of Lemma 4.1 now $\sum_e \mu_e = m|\mathcal{K}|p^{m-1} = \lambda m/p$, the conditional probabilities are $p_L = p^{|E(L)\setminus E(K)|} \le p \le \frac12$, and the conclusion becomes

$$4(1-p)\lambda m\Big[\exp\Big(-\frac{\lambda}{1-p^m} - 2\eta\Big) - \frac{\eta}{2}\Big] \le I_p^{\mathrm{s}}(f) \le 4(1-p)\lambda m \exp\Big(-\lambda + \frac{\lambda m}{\binom{n}{2}} + \frac{\lambda\eta}{2}\Big). \tag{7}$$

Theorem 8.1. *Let Γ_n satisfy the hypotheses of Theorem 4.4, where now $0 < p_n \le \frac12$ and λ_n, η_n are computed under μ_{p_n}. Then*

$$H_{p_n}(f_n) \le \Big((1+o(1))\,\frac{\log_2(1/p_n)}{1-p_n} + \log_2\Big(e\Big(1 + \frac{2\rho_n(1+\lambda_n)}{1-p_n}\Big)\Big) + o(1)\Big)\, I_{p_n}^{\mathrm{s}}(f_n),$$

where both $o(1)$ terms depend only on Λ and on upper bounds for $1/s_n$, s_n/n, $(s_n\log s_n)/m_n$ and η_n.

For $p_n = \frac12$ this is Theorem 4.4.

Proof. We follow the proof of Theorem 4.4. Put $\ell = \log_2(1/p)$ and $x = \log_2|\mathcal{K}|/m$, so that now $\log_2(1/\lambda) = m(\ell - x)$; (2) is unchanged. By (7), $I \ge I_0 = 4(1-p)\lambda m E$ with $E = e^{-\lambda}(1 + O(\eta + 2^{-m}))$. Let θ denote a nonnegative quantity of order $s/n + \eta + 2^{-m}$, not the same at each occurrence. Then

$$\frac{V}{I} \le \frac{s\log_2(en/2)}{(1-p)m}(1+\theta) \le \frac{x + s\log_2(es/2)/m}{1-p}\,(1+\theta).$$

The middle term of A is again at most $4\lambda\log_2(es\, e^{\lambda\eta/2 + o(1)}) = o(I)$. If $\lambda \ge \frac14$ then $I \ge I_0 \ge (1-p)me^{-\Lambda}(1-o(1)) \to \infty$ and $A_0/I = o(1)$. If $\lambda < \frac14$, the argument used for Theorem 4.4

gives

$$\frac{A_0}{I} \le \frac{4\lambda}{I_0}\log_2\frac{e^2 I_0}{16\lambda^2} = \frac{m(\ell-x)+\log_2\bigl(e^2(1-p)mE/4\bigr)}{(1-p)mE} \le \frac{e^\lambda(\ell-x)}{1-p}(1+\theta)+o(1).$$

In the first case $x \le \ell + \log_2\Lambda/m$. In the second $x \ge 0$, $\ell - x \ge 0$ and $(e^\lambda-1)(\ell-x) = (e^\lambda-1)\log_2(1/\lambda)/m \le 1/m$. In both cases $A/I + V/I \le \ell(1+\theta)/(1-p)+o(1)$. Finally, Lemma 4.3 and (7) give $\mathbb{E}\binom{v}{2}/I \le \bigl(1+2\rho(1+\lambda)/(1-p)\bigr)(1+o(1))$, and Proposition 2.1 completes the proof. □

For cliques, $\lambda_{n,k}(p) = \binom{n}{k}p^m$ and $\eta_{n,k}(p) = \sum_{j=2}^{k-1} t_j$ with $t_j = \binom{k}{j}\binom{n-k}{k-j}p^{m-\binom{j}{2}}$.

Lemma 8.2. *For every $\Lambda > 0$ there is $\eta^\dagger(n) \to 0$ such that $\eta_{n,k}(p) \le \eta^\dagger(n)$ whenever $3 \le k \le n$, $0 < p \le \frac12$ and $\lambda_{n,k}(p) \le \Lambda$.*

Proof. Both $\lambda_{n,k}(p)$ and every t_j increase with p. If $\lambda_{n,k}(\frac12) \le \Lambda$, then $\eta_{n,k}(p) \le \eta_{n,k}(\frac12) \le \eta^*(n)$ by Lemma 5.1. Otherwise $k < k_1(n) = O(\log n)$ by Lemma 5.1, and it suffices to take $p = p_k$, where $\lambda_{n,k}(p_k) = \Lambda$. From $(n/k)^k \le \binom{n}{k} \le (en/k)^k$,

$$\Lambda^{1/k}\frac{k}{en} \le p_k^{(k-1)/2} \le \Lambda^{1/k}\frac{k}{n}.$$

For $j = k-i$ with $1 \le i \le k/2$, the bound in the proof of Lemma 5.1 gives $t_{k-i} \le \bigl(kn\,p_k^{(3k-2)/4}\bigr)^i \le y^i$ with $y = \Lambda^{3/(2k)}k^{5/2}n^{-1/2}$, since $(3k-2)/(2k-2) \ge \frac32$. For $2 \le j \le k/2$ we have $j(j-1)/(k-1) \le j/2$, hence $p_k^{-\binom{j}{2}} \le \bigl(en/(\Lambda^{1/k}k)\bigr)^{j/2}$, and

$$t_j = \lambda\frac{\binom{k}{j}\binom{n-k}{k-j}}{\binom{n}{k}}p_k^{-\binom{j}{2}} \le \Lambda\binom{k}{j}\Bigl(\frac{k}{n-k+1}\Bigr)^j p_k^{-\binom{j}{2}} \le \frac{\Lambda}{j!}\,z^j, \qquad z = \frac{k^{3/2}(en)^{1/2}}{\Lambda^{1/(2k)}(n-k+1)}.$$

Since $k = O(\log n)$, both y and z tend to 0, and $\eta_{n,k}(p_k) \le y/(1-y) + \Lambda z^2$ once $z \le 1$. □

Theorem 8.3. *For every $\Lambda > 0$ and $\delta > 0$ there are k_0 and n_0 such that*

$$H_p(f_{n,k}) \le \Bigl((1+\delta)\frac{\log_2(1/p)}{1-p} + \log_2\Bigl(e\Bigl(1+\frac{2(1+\lambda)}{1-p}\Bigr)\Bigr) + \delta\Bigr)\,I_p^{\mathrm{s}}(f_{n,k})$$

for all $n \ge n_0$, $k_0 \le k \le n$ and $0 < p \le \frac12$ with $\lambda = \binom{n}{k}p^{\binom{k}{2}} \le \Lambda$. In the normalization of [11],

$$H_p(f_{n,k}) \le 4\Bigl((1+\delta)\log_2\frac{1}{p} + \log_2\bigl(e(3+2\lambda)\bigr) + \delta\Bigr)\,p\,I_p(f_{n,k}).$$

Proof. By Lemma 8.2, $\eta \to 0$ uniformly. For $k \le n/\log_2 n$, Theorem 8.1 applies uniformly, with $\rho = 1$, once k_0 and n_0 are large. For $k > n/\log_2 n$ we have $\lambda \le 2^n 2^{-m}$, so $\lambda n \to 0$, and the argument at the end of the proof of Theorem 1.1 applies with (7): $V/I \to 0$, $A/I \le \ell(1+\theta)/(1-p)+o(1)$ and $\mathbb{E}\binom{v}{2}/I \le 1 + 2/(1-p) + o(1)$. For the second form multiply by $I_p^{\mathrm{s}} = 4p(1-p)I_p$, and note that $(1-p)\log_2\bigl(e(1+c/(1-p))\bigr)$ decreases in p, so that it is at most $\log_2(e(1+c))$ with $c = 2(1+\lambda)$. □

Corollary 8.4. *For every $\Lambda > 0$ there is C_Λ such that $H_p(f_{n,k}) \le C_\Lambda\,p\log_2(1/p)\,I_p(f_{n,k})$ for all $3 \le k \le n$ and all $0 < p \le \frac12$ with $\lambda_{n,k}(p) \le \Lambda$. Moreover, as $p \to 0$ and $k \to \infty$ with $\lambda \le \Lambda$,*

$$\limsup \frac{H_p(f_{n,k})}{p\log_2(1/p)\,I_p(f_{n,k})} \le 4,$$

and 4 cannot be replaced by a smaller constant in the conjecture of [11]: for $f(x) = x_1$ the ratio equals $h(4p(1-p))/(p\log_2(1/p))$, which tends to 4.

Proof. Take $\delta = 1$ in Theorem 8.3. Since $\log_2(1/p) \geq 1$, the ratio is at most $4(3+\log_2(e(3+2\Lambda)))$ when $n \geq n_0$ and $k \geq k_0$. In the remaining cases $\log_2 N = O_\Lambda(\log_2(1/p))$: if $k < k_0$, then $p^{(k-1)/2} \leq \Lambda^{1/k} k/n$ gives $\log_2(1/p) \geq \log_2 n/k_0$ for n large, and otherwise n is bounded and $\log_2(1/p) \geq 1$. In these cases [11, Claim 2.5], which states $H_p(f) \leq c\, p(1-p) \log N\, I_p(f)$ for every f of N variables, gives the bound. The upper limit follows from Theorem 8.3 by letting $\delta \to 0$. □

The local properties of Section 6 behave in the same way. The edge-isoperimetric inequality is replaced by the following biased version for increasing sets.

Lemma 8.5. *Let $0 < p < 1$ and let $A \subseteq \{0,1\}^N$ be increasing, with $\mu = \mu_p(A) > 0$. Then $p\, I_p(A) \geq \mu \log_p \mu$, where $I_p(A) = \sum_i \mathbb{P}_p[\mathbf{1}_A(x) \neq \mathbf{1}_A(x \oplus e_i)]$, with equality when $A = \{x : x_i = 1 \text{ for } i \in T\}$.*

Proof. Induction on N. Split A by the last coordinate into increasing sets $A_0 \subseteq A_1 \subseteq \{0,1\}^{N-1}$ with measures $a \leq b$. Then $\mu = (1-p)a + pb$ and $I_p(A) = (1-p)I_p(A_0) + pI_p(A_1) + (b-a)$. With $\varphi(x) = x \log_p x$ and the induction hypothesis it suffices to show that $D(a,b) = (1-p)\varphi(a) + p\varphi(b) + p(b-a) - \varphi((1-p)a + pb)$ is nonnegative for $0 \leq a \leq b \leq 1$. This holds because $D(a,a) = 0$ and $\partial D/\partial b = p \log_p\big(pb/((1-p)a + pb)\big) \geq 0$. □

Remark 8.6 (Local properties under μ_p). Let $0 < p_n \leq \frac{1}{2}$, let B_n be as in Section 6 with π_n, λ_n, η_n computed under μ_{p_n}, and put $\iota_n = I_{p_n}(B_n)/\pi_n$. Suppose that $\lambda_n \leq \Lambda$, $\eta_n \to 0$ and $k_n \log k_n + k_n \log_2(1/p_n) = o(\log_2(1/\pi_n))$. Then

$$H_{p_n}(f_n) \leq \Big((1+o(1))\frac{\log_2(1/p_n)}{1-p_n} + \log_2\Big(e\Big(1 + \frac{(1+\lambda_n)k_n^2}{p_n(1-p_n)\iota_n}\Big)\Big) + o(1)\Big) I^{\mathrm{s}}_{p_n}(f_n).$$

The proof is that of Theorem 6.1. The influence bound (6) becomes $I^{\mathrm{s}}_p \geq 4p(1-p)q\lambda\iota(1-\theta)$. Lemma 8.5 gives $p\,\iota \geq \log_2(1/\pi)/\log_2(1/p)$, which turns the estimate for V/I into $V/I^{\mathrm{s}}_p \leq \log_2(1/p)(1+\theta)/(1-p)$. The hypothesis on $k \log_2(1/p)$ makes the middle term of A negligible. For $B = \{K_k\}$ we have $p\,\iota = m$, and the bound is that of Theorem 8.1.

Remark 8.7 (Finite n under μ_p). The bound $B(n,k,p)$ for H_p/I^{s}_p obtained by evaluating the inequalities of this section exactly, as in Remark 9.1, is shown below, together with the bound of Theorem 8.3 for $\delta = 0$ ("limit") and with B in the normalization of [11]; at $p = \frac{1}{2}$ the computation reproduces the values of Remark 9.1. For small k the bound stays above the limit, because the terms of order $(\log k)/k$ in the proof do not vanish. For $n \leq 6$, every $3 \leq k \leq n$ and $p \in \{0.5, 0.3, 0.1, 0.03\}$, the spectral entropy was computed exactly under μ_p (except for six cases in which $\mathbb{P}[S \neq \emptyset]$ is below double precision): Lemmas 3.1 and 3.2 hold with ratios at most 1 and 0.38, H_p/I^{s}_p lies below the right-hand side of Proposition 2.1, which lies below B, and the ratio $H_p/(p\log_2(1/p)I_p)$ lies between 4.02 and 4.08 at $p = 0.03$ and between 4.45 and 4.63 at $p = 0.1$.

p	n	k	λ	$B(n,k,p)$	limit	$4(1-p)B/\log_2(1/p)$
2^{-2}	2^{20}	18	2.813	7.53	7.59	11.29
2^{-4}	2^{24}	12	0.035	7.64	7.39	7.17
2^{-8}	2^{20}	6	0.0014	11.83	11.06	5.89
2^{-16}	2^{40}	6	0.0014	19.79	19.03	4.95

9. Remarks

Remark 9.1 (Finite n). The inequalities of Section 4 can be evaluated exactly for given n and k: Proposition 2.1 with Lemmas 3.1 and 3.2, Janson's inequality with its second form $e^{-\mu^2/(2\Delta)}$

when $\Delta \geq \mu$, the lower bound $\prod_K (1-2^{-m})$ for $\mathbb{P}[f=0]$, and the Bonferroni and second-moment lower bounds for $\mathbb{P}[e \text{ pivotal}]$. The resulting upper bound $B(n,k)$ for H/I, computed with 60 significant digits, is shown below for some $n = 2^j$. For $n = 2^j$ with $12 \leq j \leq 37$ it is at most 8.8 at every k with $2^{-12} \leq \lambda \leq 11$; at the three larger values of λ in this range it is 9.85 ($n = 2^{22}$, $\lambda = 25.9$), 11.0 ($n = 2^{31}$, $\lambda = 61.2$) and 19.9 ($n = 2^{15}$, $\lambda = 30.9$). For $n \leq 2^{11}$ and λ above about 2 the Janson factor and the lower bound for I are too weak, and the bound is large. For $n \leq 7$ the spectral entropy can be computed exactly: over all $2 \leq k \leq n \leq 7$ the ratio H/I lies between 2.90 (triangles on 4 vertices) and 4.35 (triangles on 7 vertices), the right-hand side of Proposition 2.1 is at most $6.47\,I$, and $B(n,k)$ lies above both in every case. Far below the threshold the two small terms trade off as in the proof of Theorem 4.4: for $n = 2^{16}, 2^{20}, 2^{30}, 2^{40}$ and every k with $2^{-3600} \leq \lambda \leq 1$, the computed bounds for A/I and V/I add up to at most 2.14, decreasing towards 2 as k grows, and $B(n,k)$ exceeds $2 + \log_2(e(5+4\lambda))$ by at most 0.16.

n	k	λ	$B(n,k)$	$2+\log_2(e(5+4\lambda))$
2^{12}	19	1.136	8.47	6.70
2^{17}	28	1.036	6.75	6.64
2^{24}	41	0.699	6.41	6.41
2^{33}	58	1.576	6.93	6.94
2^{32}	56	10.18	8.75	8.96
2^{36}	64	0.0039	5.79	5.77

The bound can lie below the limit value because at finite n the vertex term V/I is below its limit 2; at $\lambda = 10.18$ it is 1.77.

Remark 9.2 (Graph properties). For a graph property f the spectral distribution is invariant under permutations of the vertices, so conditionally on v the set $V(S)$ is uniformly distributed among the v-subsets of $[n]$, and $H(V(S)) = H(v) + \mathbb{E}\log_2 \binom{n}{v}$ exactly. Here $H(v) \leq h(\varepsilon) + \varepsilon \log_2(e\,\mathbb{E}v/\varepsilon)$, $\mathbb{E}v \leq 2I$ because $v \leq 2|S|$, and $h(\varepsilon) = O(I)$ by the edge-isoperimetric inequality, so $H(v) = O(I)$. Hence a class of graph properties satisfies the conjecture if and only if

$$\mathbb{E}\log_2 \binom{n}{v} = O(I) \qquad \text{and} \qquad H\big(S \mid V(S)\big) = O(I)$$

on the class. The second condition holds when $\mathbb{E}\binom{v}{2} = O(I)$, that is, when the spectral sample is dense in its vertex set on average, by the proof of Proposition 2.1.

Remark 9.3 (Hypergraphs). For a Boolean function of the r-element subsets of $[n]$, Proposition 2.1 holds with $V(S)$ the union of the r-sets in S and $\binom{v}{r}$ in place of $\binom{v}{2}$, and Lemmas 3.1, 4.1 and 4.2 hold with the same proofs for copies of an r-uniform hypergraph, reading hyperedge for edge. Lemma 3.2 has the following r-set version, proved in the same way. Let $R = \{u_1, \dots, u_r\}$, let a_J ($\emptyset \neq J \subseteq [r]$) be the indicator that G contains a minterm touching u_i exactly for $i \in J$, and g that of a minterm touching none of them. Then

$$\mathbb{P}[R \subseteq V(S)] \leq \mathbb{P}[R \in S] + 4\Big(\sum_{\mathcal{J}} \big(\mathbb{E}\big[\textstyle\prod_{J\in\mathcal{J}} a_J\big]\mathbb{P}[g=0]\big)^{1/2}\Big)^2,$$

the sum over the families $\mathcal{J}$ of nonempty subsets of $[r]$ that cover $[r]$: with A_i the hyperedges containing u_i other than R, the operator $\prod_i (1 - \mathbb{E}_{A_i})$ annihilates every product of the a_J that does not cover $[r]$, and Minkowski's inequality over the fibres and the Harris inequality finish as before. For $r = 3$ this was checked in exact arithmetic on the complete 3-graphs $K_4^{(3)}$ and $K_5^{(3)}$ with $n \leq 6$, where the ratio of the two sides before the Harris step is at most 0.92. For the complete r-graph at its threshold, the products over covers with $t \geq 3$ members require a bound

on the t-th factorial moments of the number of copies present, for $t < 2^r$. With such a bound the proof of Theorem 4.4 gives $H \le C_r I$ in the window $\Lambda^{-1} \le \lambda \le \Lambda$. For $r = 2$ only pairs of copies enter, which is why Theorem 1.1 needs no such input.

Remark 9.4. The width and read bounds of [18, 19] come from an induction on coordinates, in which one variable is restricted at a time and the entropy lost is compared with its influence. Here an edge has influence of order $(\log n/n)^2$, and its restriction produces a function that is no longer a clique property. The coding of Proposition 2.1 uses the vertices instead. For a general monotone function the analogue would be a small family of sets of variables, playing the role of the vertex stars, that covers the spectral sample economically; for the clique property the stars are the natural choice because the minterms are determined by their vertex sets.

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Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies

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